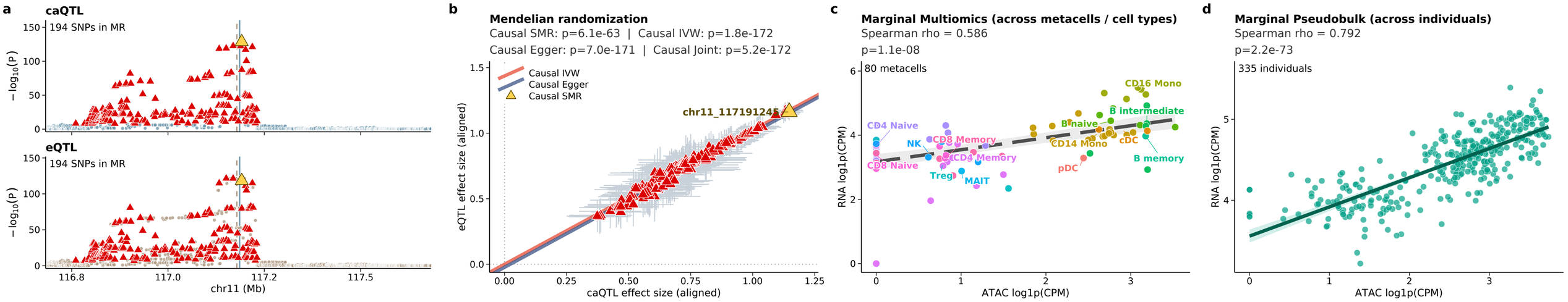
# Example peak-gene pairs

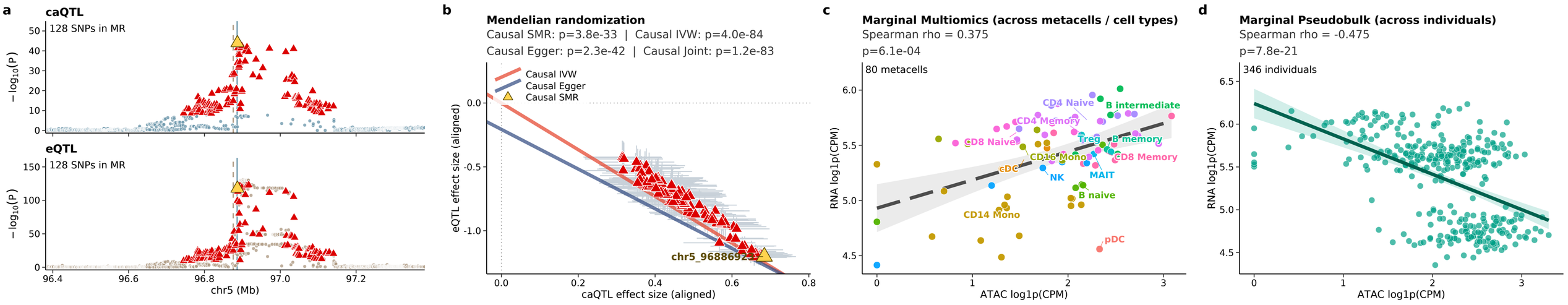
## SIDT2 | chr11:117185715-117186216 (Bn_TCL1A)

peak-TSS distance: 7.2 kb | 194 SNPs in MR



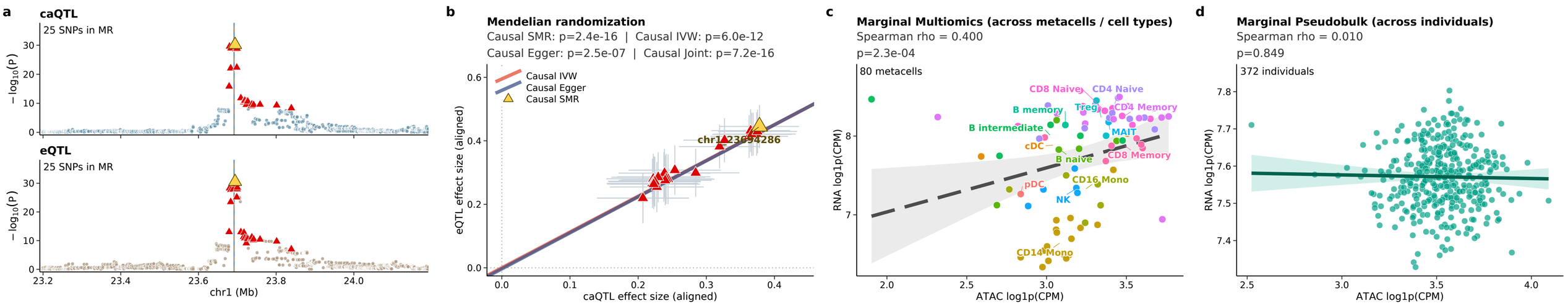

## ERAP2 | chr5:96885894-96886395 (CD8_Tn_CCR7)

peak-TSS distance: 10.2 kb | 128 SNPs in MR



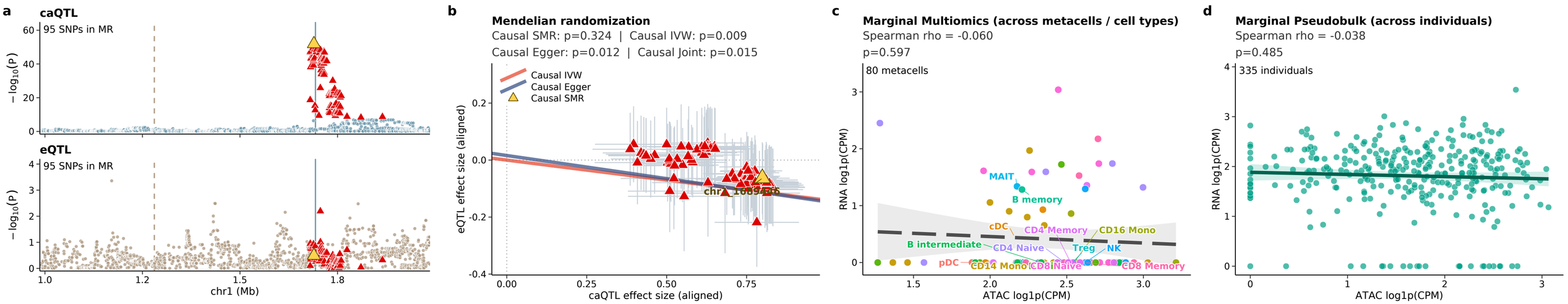

## RPL11 | chr1:23691494-23691995 (CD4_Tn_CCR7)

peak-TSS distance: 2 bp | 25 SNPs in MR




## SCNN1D | chr1:1693193-1693694 (Bn_TCL1A)

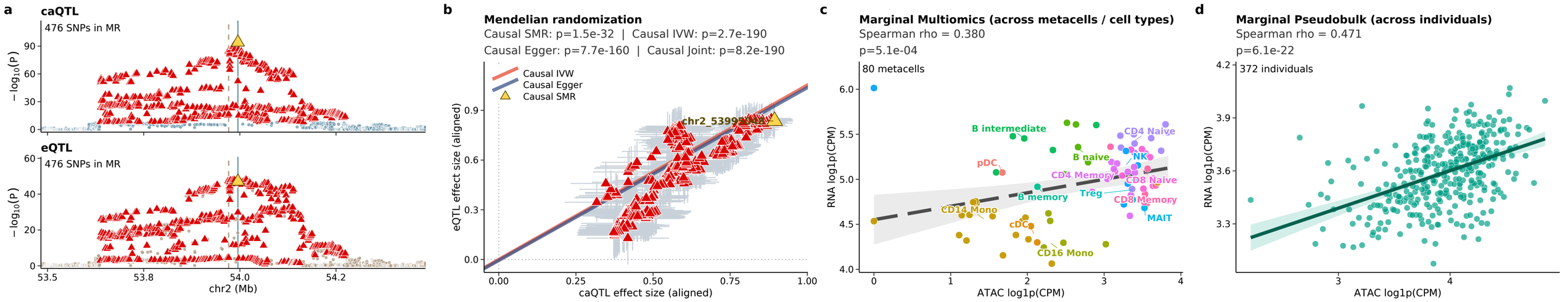
peak-TSS distance: 413.0 kb | 95 SNPs in MR



# Example peak-gene pairs by type (set 1)

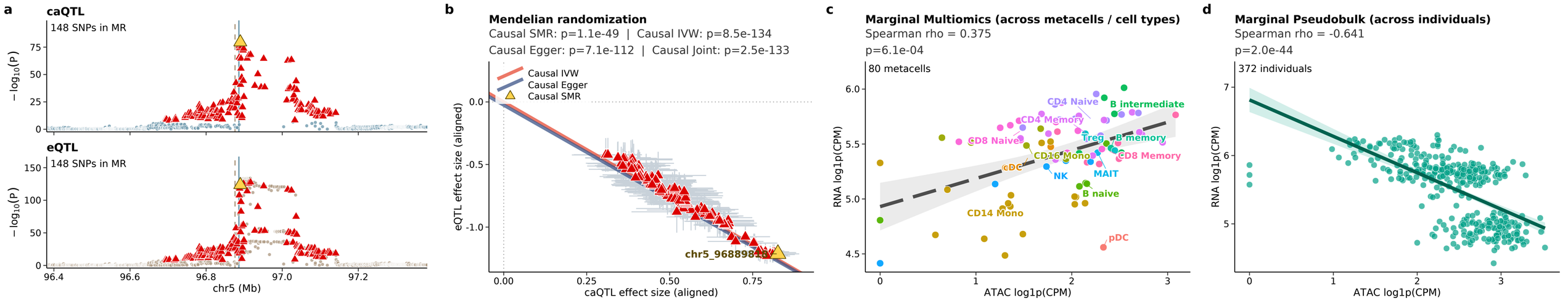
## ACYP2 | chr2:53994688-53995189 (CD4_Tn_CCR7)

Type 1: All methods significant | peak-TSS distance: 24.1 kb | 476 SNPs in MR



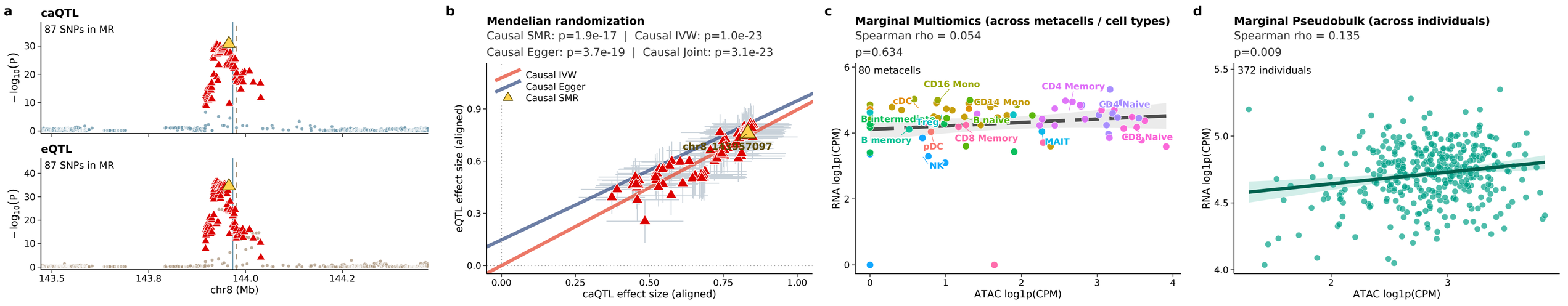

## ERAP2 | chr5:96885894-96886395 (CD4_Tn_CCR7)

Type 2: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 10.2 kb | 148 SNPs in MR



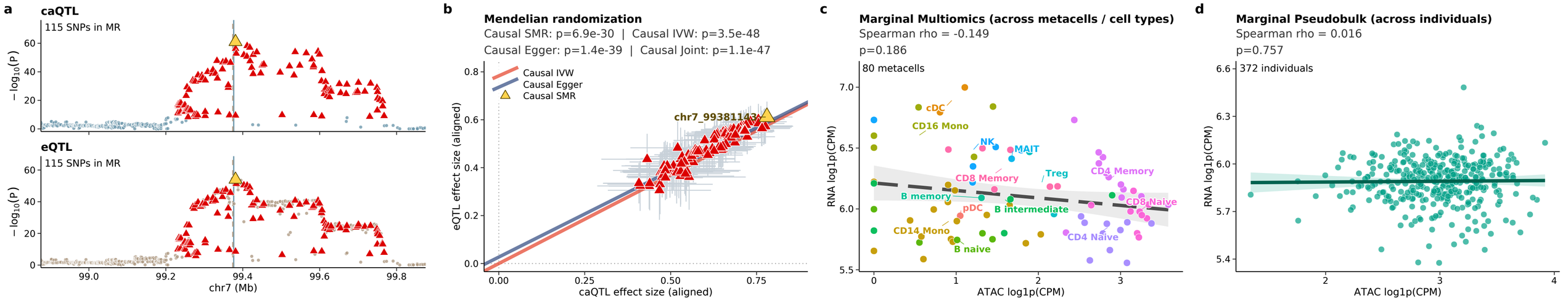

## PLEC | chr8:143966529-143967030 (CD4_Tn_CCR7)

Type 3: Opposite direction (MR vs. marginal) | peak-TSS distance: 10.0 kb | 87 SNPs in MR




## ARPC1B | chr7:99376204-99376705 (CD4_Tn_CCR7)

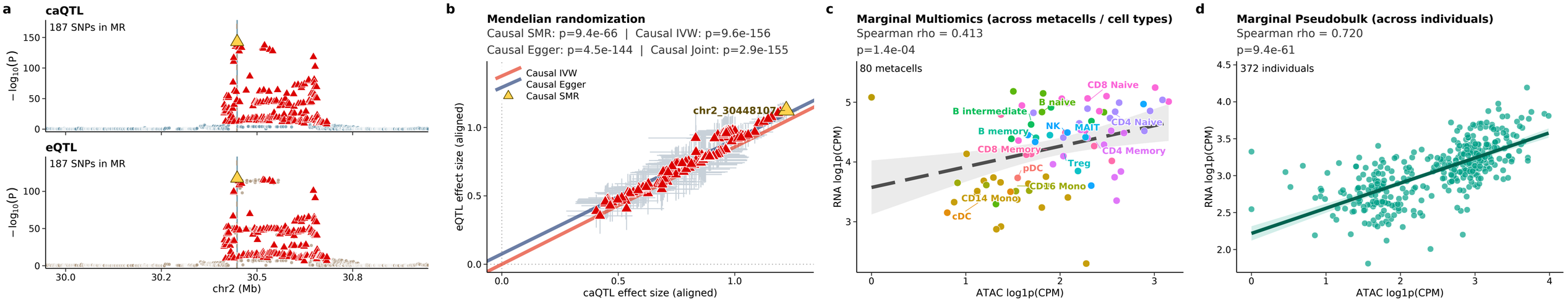
Type 4: Significant only in MR | peak-TSS distance: 2.2 kb | 115 SNPs in MR



# Example peak-gene pairs by type (set 2)

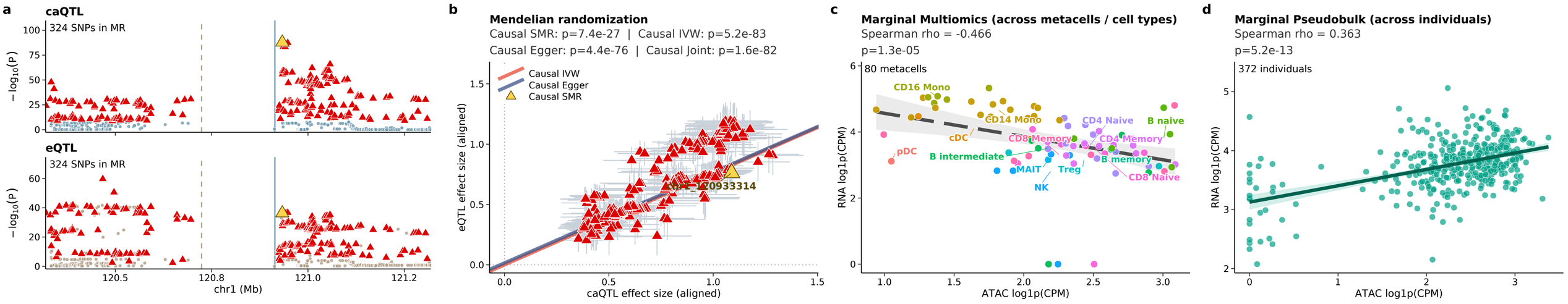
## LCLAT1 | chr2:30447755-30448256 (CD4_Tn_CCR7)

Type 1: All methods significant | peak-TSS distance: 780 bp | 187 SNPs in MR



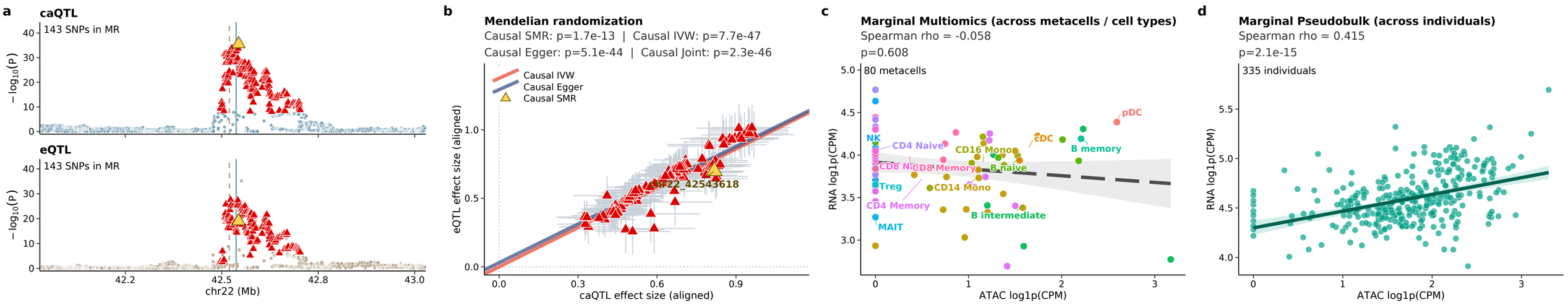

## NBPF26 | chr1:120913924-120914425 (CD4_Tn_CCR7)

Type 2: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 190.2 kb | 324 SNPs in MR



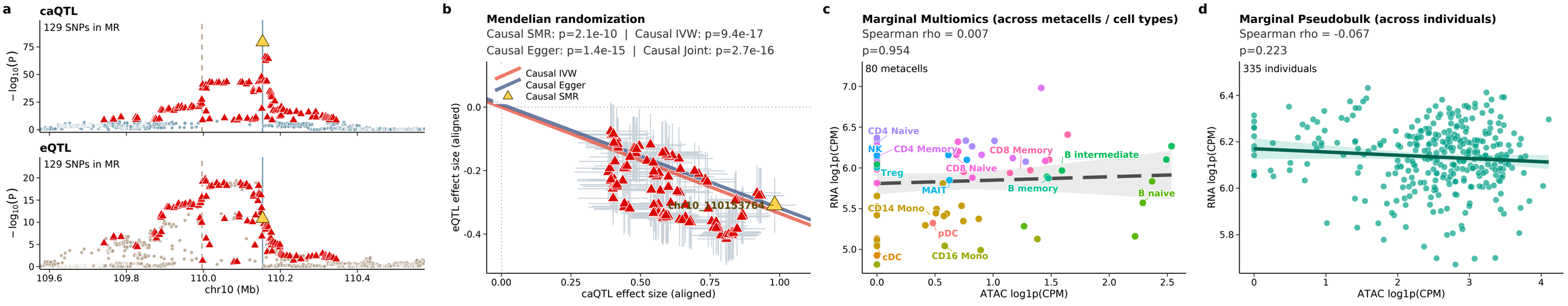

## RRP7A | chr22:42537036-42537537 (Bn_TCL1A)

Type 3: Opposite direction (MR vs. marginal) | peak-TSS distance: 17.5 kb | 143 SNPs in MR




## ADD3 | chr10:110153435-110153936 (Bn_TCL1A)

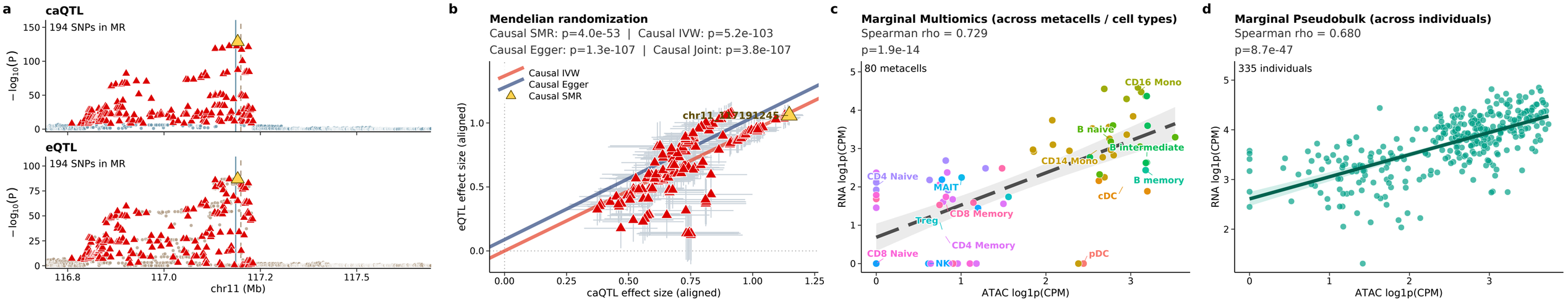
Type 4: Significant only in MR | peak-TSS distance: 157.3 kb | 129 SNPs in MR



# Example peak-gene pairs by type (set 3)

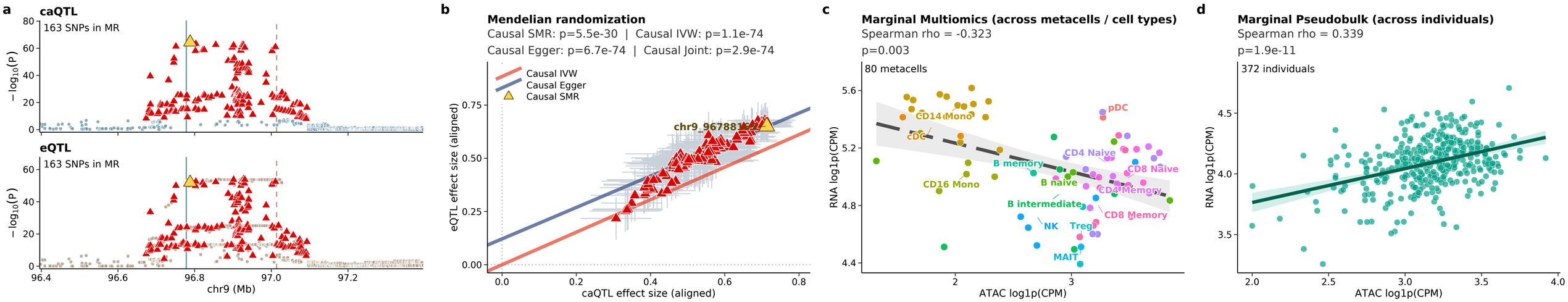
## TAGLN | chr11:117185715-117186216 (Bn_TCL1A)

Type 1: All methods significant | peak-TSS distance: 13.4 kb | 194 SNPs in MR



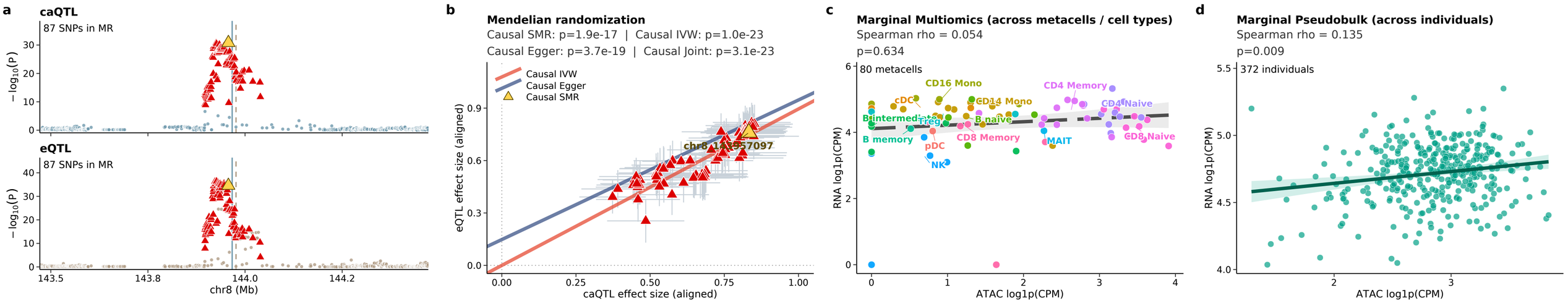

## MFSD14C | chr9:96777893-96778394 (CD4_Tn_CCR7)

Type 2: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 235.6 kb | 163 SNPs in MR



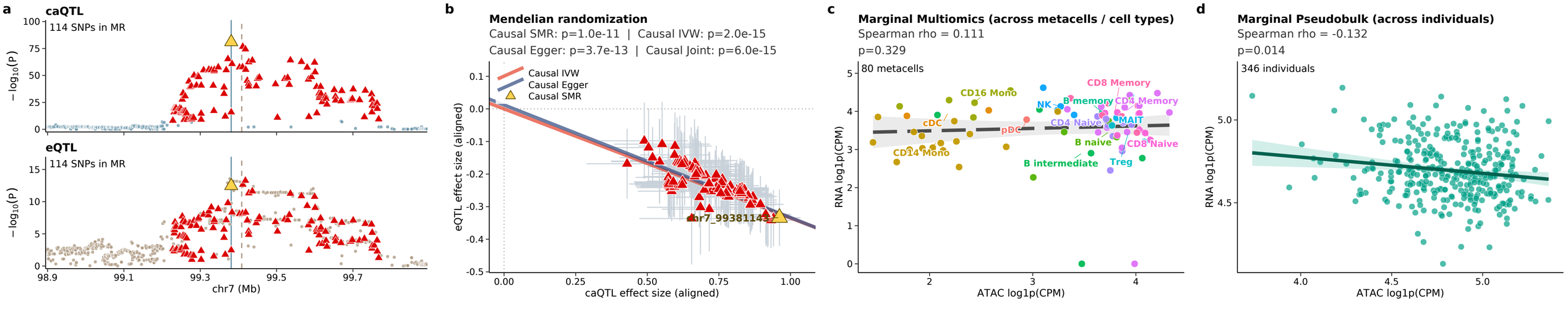

## PLEC | chr8:143966529-143967030 (CD4_Tn_CCR7)

Type 3: Opposite direction (MR vs. marginal) | peak-TSS distance: 10.0 kb | 87 SNPs in MR




## BUD31 | chr7:99380850-99381351 (CD8_Tn_CCR7)

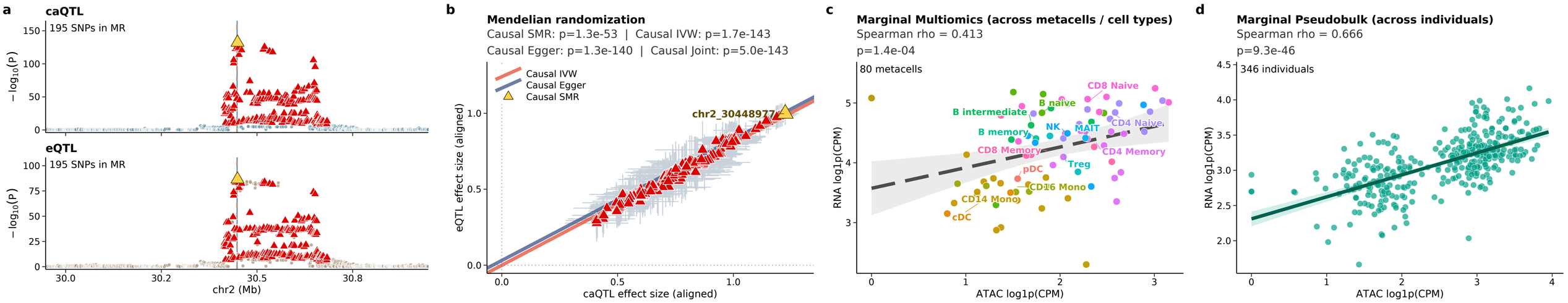
Type 4: Significant only in MR | peak-TSS distance: 27.5 kb | 114 SNPs in MR



# Example peak-gene pairs by type (set 4)

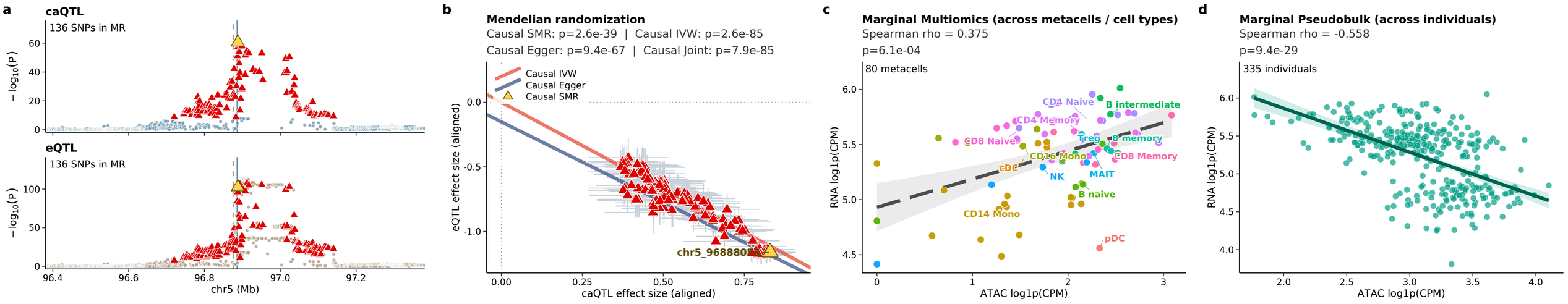
## LCLAT1 | chr2:30447755-30448256 (CD8_Tn_CCR7)

Type 1: All methods significant | peak-TSS distance: 780 bp | 195 SNPs in MR



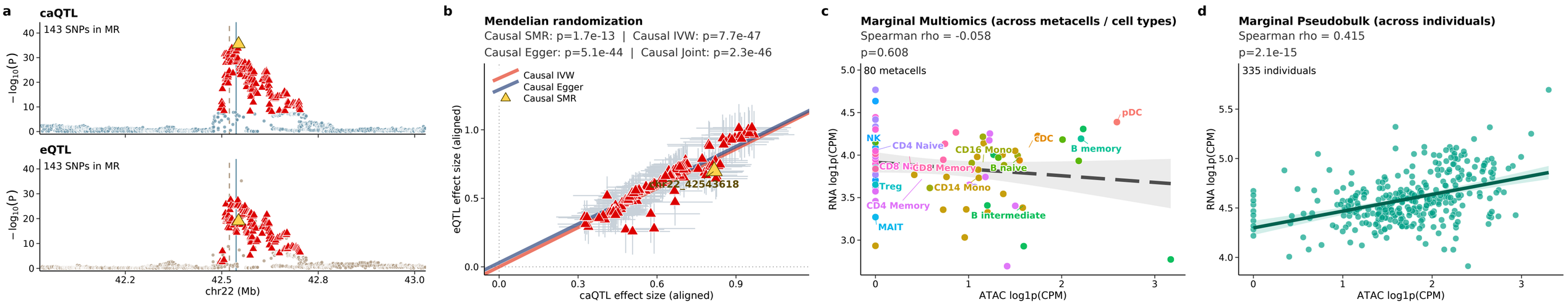

## ERAP2 | chr5:96885894-96886395 (Bn_TCL1A)

Type 2: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 10.2 kb | 136 SNPs in MR



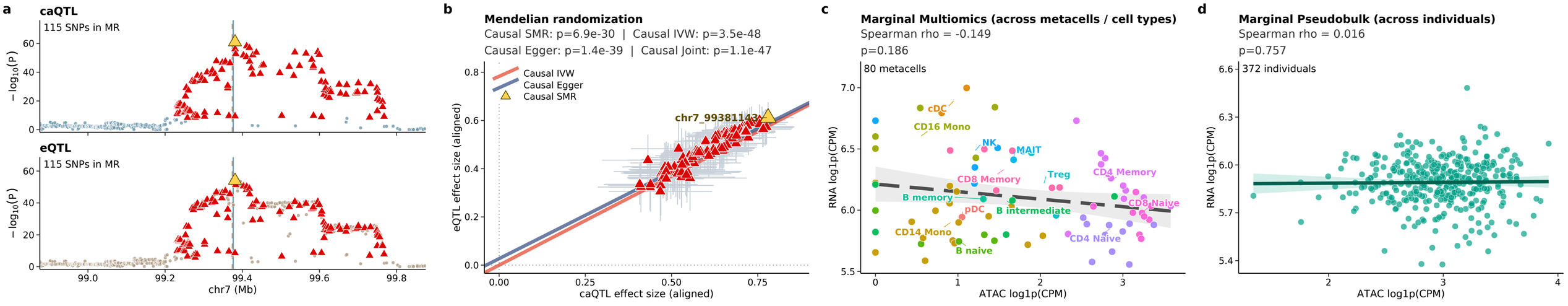

## RRP7A | chr22:42537036-42537537 (Bn_TCL1A)

Type 3: Opposite direction (MR vs. marginal) | peak-TSS distance: 17.5 kb | 143 SNPs in MR




## ARPC1B | chr7:99376204-99376705 (CD4_Tn_CCR7)

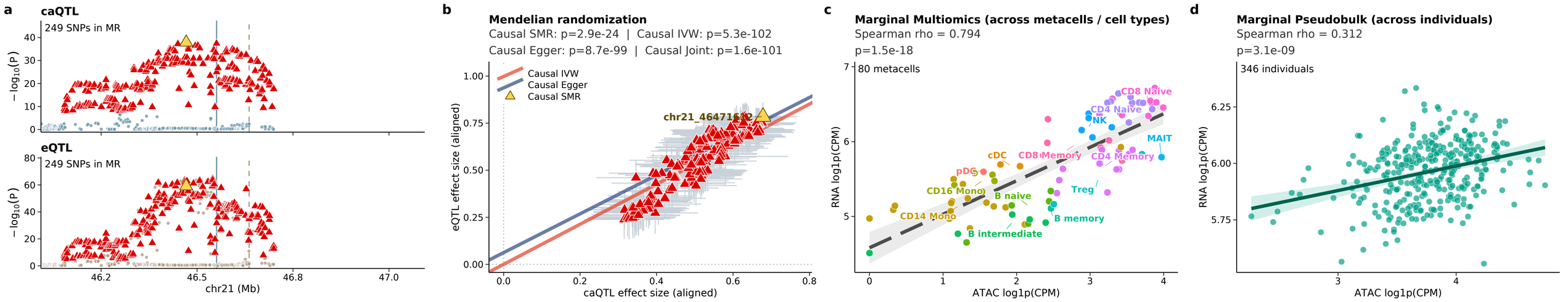
Type 4: Significant only in MR | peak-TSS distance: 2.2 kb | 115 SNPs in MR



# Example peak-gene pairs by type (set 5)

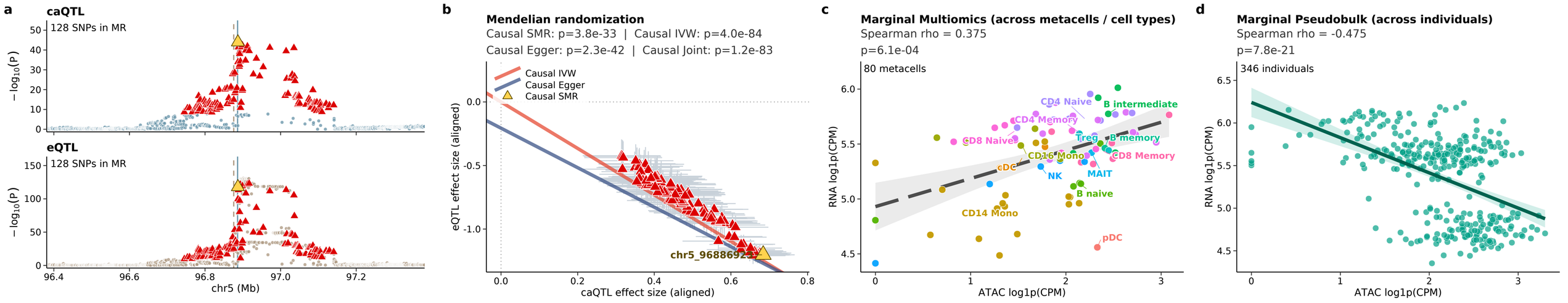
## PRMT2 | chr21:46550539-46551040 (CD8_Tn_CCR7)

Type 1: All methods significant | peak-TSS distance: 84.8 kb | 249 SNPs in MR



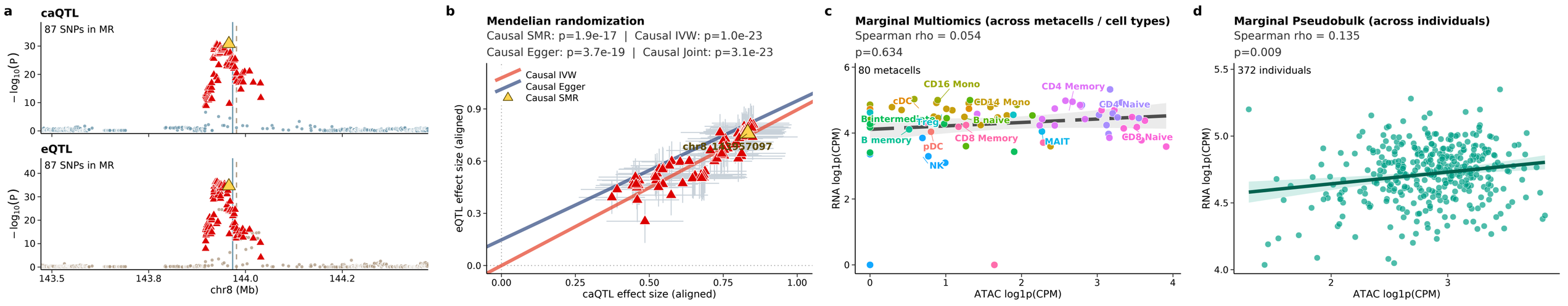

## ERAP2 | chr5:96885894-96886395 (CD8_Tn_CCR7)

Type 2: Not significant in Marginal Multiomics / Pseudobulk | peak-TSS distance: 10.2 kb | 128 SNPs in MR




## PLEC | chr8:143966529-143967030 (CD4_Tn_CCR7)

Type 3: Opposite direction (MR vs. marginal) | peak-TSS distance: 10.0 kb | 87 SNPs in MR




## ADD3 | chr10:110153435-110153936 (Bn_TCL1A)

Type 4: Significant only in MR | peak-TSS distance: 157.3 kb | 129 SNPs in MR

